

## Histology

### سوال 1-

**Formation of barrier in CNS and PNS is the function of:**

Oligodendrocyte- Schwann cell

Astrocyte- Satellite cell

Oligodendrocyte- Myelin sheat

✓ Astrocyte- Perineurium

### سوال 2-

**Which type of junctions is for adhesion of cell to basement membrane?**

Tight J

Desmosomes

Gap J

✓ Hemidesmosomes

### سوال 3-

**This type of neuron consist of many dendrites and one true axon , and are more:**

Apolar

Bipolar

✓ Multipolar

Unipolar or pseudo unipolar

## سوال 4-

Which layer of the epidermis has cells which have keratohyaline granules?

Stratum corneum

Stratum spinosum

✓ Stratum granulosum

Stratum basale

## سوال 5- heat production is the most important function in which tissue?

dermis

areolar tissue

✓ brown adipose tissue

tendon

## سوال 6-

Axons have all organelles except:

Mitochondria

✓ Ribosomes

Lysosomes

Peroxisome

## سوال 7-

Which cell is sensory reception between epidermal cells?

melanocyte

lymphocyte

keratinocyte



merkel

## سوال 8-

In which muscle type(s) nucleus located in center part?

smooth muscle and skeletal muscle

smooth muscle and striated muscle



cardiac muscle and smooth muscle

skeletal muscle and cardiac muscle

## 9- In muscular tissue.....

Muscle cells attached together by junctions

A band include just thin filaments



H band include just thick filaments

M band include just Kratin kinase filaments

10- سوال which one is dense regular connective tissue?

Dermis of skin

Adipose tissue

✓ Tendon

Areolar tissue

11- سوال Which of the following histological structures has Mesenchymal Like cells?

Endosteum

Periosteum

Volkmann Canal

✓ Perichondrium

12- سوال

What type of cellular junctions do osteocyte canals of bone tissue connect to each other?

Desmosome

Adherens

Tight

✓ Gap junction

13- سوال In which of the following is Observed type II collagen ?

Matrix

Perichondrium

Elastic Cartilage

✓ Hyalin Cartilage

14- سوال

Which one makes Nissl bodies?

Smooth Endoplasmic reticulum

Mitochondria

✓ Ribosomes and Rough Endoplasmic reticulum

Synaptic vesicles

15- سوال Which of the following Is Observed in the histological structure

Metaphysis Plate?

Conderoclast

Osteocyte

✓ Chonderocyte

Osteoclast

## سوال 16-

What do you call the simple squamous epithelium that lines the serous membranes?

- Epithelioid
- Pseudostratified
- Endothelium
- ✓ Mesothelium

## سوال 17- in which cell we have major basic protein?

- macrophage
- ✓ Eosinophil
- lymphocyte
- monocyte

## سوال 18- structure of framework of spleen and lymphnodes are composed of which fibers?

- ✓ reticular
- elastic
- elaunin
- collagen bundles

## سوال 19-

**Spinal central canal lined by:**

Schwann cell

Microglia

Oligodendrocyte

✓ Ependymal cell

## سوال 20-

**Which one is located on the dendrites?**

Golgi complex

✓ Dendritic spines

Nucleus

Nissl bodies

## سوال 21- In which of the following is Observed Osteoclast ?

Periosteum

Nothing

Pericondrium

✓ Endosteum

## سوال 22-

Which cell is a macrophage found in the skin?

- Microglia
- ✓ Langerhans cell
- Dust cell
- Histiocyte

## سوال 23-

in which of the cartilage tissue Isnt Observed perichondrium?

- Larynex Cartilage Tissue
- Sulcus Skull Cartilage Tissue
- ✓ Joint Cartilage Tissue
- Pinna Cartilage Tissue

24- سوال Red blood cell is ..... because it has ..... In the wall.

- ✓ flexible-spectrin
- fragile-actin
- fragile-spectrin
- flexible-fibrinogen

## سوال -25

Which type of exocrine glands lose whole of cell bodies during secretion?

- Apocrine
- endocrine
- ✓ Holocrine
- Merocrine

## سوال -26 which cell has fewest number in the normal blood smear?

- monocyte
- Eosinophil
- Neutrophil
- ✓ Basophil

## سوال -27

What is the covering sheath of a muscle?

- Sarcolemma
- ✓ Epimysium
- Endomysium
- Perimysium

## سوال -28

What type of muscle has branching cells?

Striated and Smooth



Cardiac

Skeletal

Smooth

## سوال -29

Which type of regulatory protein located in M line of the skeletal muscle?

Neboulin

actinin



Myomesin

Titin

## سوال -30

Which of the following Fibers is observed most in the elastic cartilage perichondrium?

Reticular

Elastic

Oxitalan



Collagen I

## سوال - 31

Which of the following terms is applied to collection of neuron cell bodies (Soma) in the peripheral nervous system?

White matter

Gray matter

Nucleus

✓ Ganglion

## سوال - 32 Russel bodies are characteristic for which cell?

Adipocyte

mastcell

kupfer

✓ plasmacell

## سوال - 33

What type of cells makes up the most part of skin hypodermis?

Melanocyte

Fibroblast

✓ Adipocyte

Macrophage

34- سوال following diapedesis only type of leukocytes can return from the tissues back to the blood. Which one is this?

- ✓ Lymphocyte
- basophil
- neutrophil
- monocyte

35- سوال

Identify the INCORRECT statement

- In pseudostratified epithelia all cells are in contact with the basement membrane
- ✓ Epithelia are classified by the shape of cell layers.
- Microvilli increase the surface area of the epithelia cells.
- The basement membrane consists of two sub layers: the reticular layer and the basal lamina.

36-

سوال

About Unmyelinated PNS axons, which choice is correct?

- Oligodendrocyte cell starts to envelop multiple axons
- ✓ Schwann cell starts to envelop multiple axons
- Schwann cell starts to envelop one axon
- Oligodendrocyte cell starts to envelop one axon

Physiology

**37- Which transmitter is released from skeletal muscle fibers into the neuro-muscular junction?**

- Epinephrine
- ✓ Acetylcholine
- GABA
- Norepinephrine

**38- Which of the following is not a type of “muscle remodeling”?**

- ✓ Muscle fatigue
- Muscle Atrophy
- Muscle Hypertrophy
- Hyperplasia

**39- During discharge of a presynaptic action potential  $\text{Ca}^{2+}$  enters the presynaptic terminal through ..... at the active zone.**

- ✓ Voltage-gated  $\text{Ca}^{2+}$  channels
- Voltage-gated  $\text{Na}^{+}$  channels
- $\text{Na}^{+}$ - $\text{Ca}^{2+}$  channels
- Voltage-gated  $\text{K}^{+}$  channels

**40- Which factor is initiator in the skeletal muscle contraction?**

- ✓ Calcium ions
- ATP
- Troponin
- Tropomyosin

**41- In which step of contraction cycle, the cycle is stopped during rigor mortis?**

- Step 4
- Step 1
- Step 2
- ✓ Step 3

**42- Which of the following suggests a rudimentary analog of the transverse tubule system in smooth muscle?**

- Synapsin
- ✓ Caveolae
- Troponin complex
- Myosin light chain kinase

**43- Which type of receptors produce slower synaptic actions lasting seconds to minutes?**

- ✓ Metabotropic receptors
- Ionotropic receptors
- Receptor-channels
- Ligand-gated channels

**44- Which of the following maintains "muscle tautness"?**

- ✓ Low rate of impulses coming from the spinal cord
- Depletion of muscle glycogen
- Decreased blood flow
- Loss of ATP

**45- Which of the following covers active sites in skeletal muscle fibers?**

- Myosin filaments
- G actin molecules
- F actin strands
- ✓ Tropomyosine strands

**46- Which of the following shows the electrical synapses property?**

- Chemical transmitter
- Synaptic delay
- ✓ Bidirectional transmission
- Presynaptic vesicles

**47- Which of the following best describes the significant difference between skeletal muscle and large nerve fibers?**

- ✓ Velocity of conduction
- Generation of action potential
- Voltage gated sodium channels presense
- Resting membrane potential to an approximately equal extent

48- Which of the following is “regulatory protein” in smooth muscle contraction?

Troponin

Tropomyosin

Actin

✓ Calmodulin

49- What is the importance of chemical synapses?

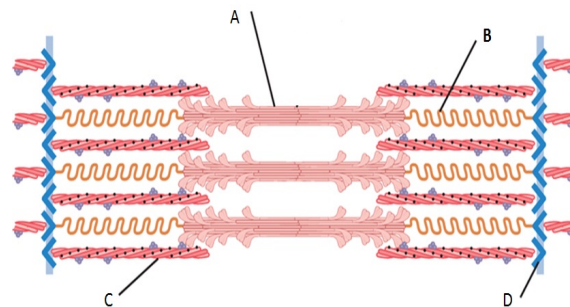
They allow the synchronous firing of interconnected cells

Gap junctions faster the conduction velocity

It is extremely rapid

✓ They can amplify signals.

50- In the figure, which of the following points out “titin protein”?



C

✓ B

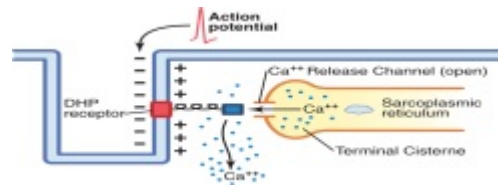
A

D

51- Which of the following is a unitary smooth muscle?

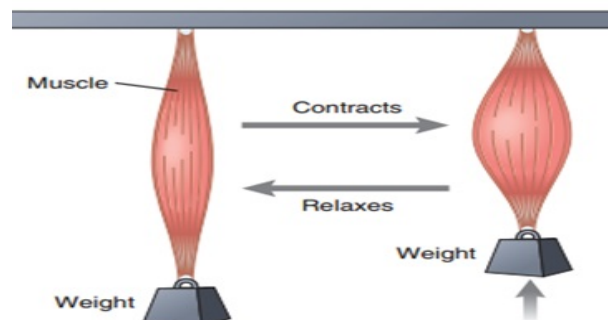
- Ureter
- ✓ Ciliary muscle of the eye
- Gastrointestinal tract
- Bile duct

52- What does this figure show?



- ✓ Excitation-contraction coupling in skeletal muscle
- Excitation- contraction coupling in smooth muscle
- Action potential generation
- Transmitter release in nerve terminal

53- Which phenomenon is shown in this figure?



Isometric contraction



Isotonic contraction

Tetanus

staircase effect

54- Action potentials spread to the interior of the skeletal muscle fiber by way of ..... .

longitudinal tubules

reticulum endoplasmic

terminal cisternae



transverse tubules

**55- Myosine phosphatase is involved in ..... .**

- ☐ Smooth muscle contraction
- ☒ Smooth muscle relaxation
- ☐ Skeletal muscle relaxation
- ☐ Skeletal muscle contraction

**56- What is the importance of “subneural clefts” in neuro-muscular junction?**

- ☐ It forms a complex of branching nerve terminals that invaginate into the surface of the muscle fiber.
- ☐ It allows the vesicles to move to the active zone of the presynaptic neural membrane.
- ☒ It greatly increases the surface area at which the synaptic transmitter can act.
- ☐ It is the space between the nerve terminal and the fiber membrane.

**57- In which type of following muscle, the source of ca ions needed for contraction is just from sarcoplasmic reticulum?**

- ☒ Soleus muscle
- ☐ Iris muscle of the eye
- ☐ Uterus
- ☐ Blood vessels

**58- Which factor causes detachment of the myosin head from the actin filaments in skeletal muscle fibers?.**

- ☒ Binding of new ATP
- ☐ ATPase activity of the myosin head
- ☐ Release of the ADP and phosphate ion that were previously attached to the head
- ☐ Tilt of the head toward the arm

## Embryology

### 59- The primitive streak

- ✓ is derived from epiblast
- .first appears in the roof of the amniotic cavity
- gives rise to the allantois
- none of the above

### 60- Select the correct options below for the structures formed by sclerotome

- ✓ .vertebra
- .dermis
- . limb bud cartilage
- .intervertebral disc

### 61- Syncytiotrophoblast

- ✓ secretes chorionic gonadotropin
- gives rise to the decidua basalis
- begins to develop on the 4th day post fertilization
- degenerates once cytotrophoblast takes over its function

**62- Primordial Germ cells are derived of:**

- ✓ .Epiblast
- endodermal cells
- .mesoderm
- hypoblast

**63- Which of the following may be associated with polyhydramnios (i.e., may cause polyhydramnios or may result from it)**

- duodenal atresia
- anencephaly
- proximal jejunal atresia
- ✓ all of the above

**64- In which kind of twin is twin-twin transfusion syndrome the greatest threat to proper development**

- ✓ Monochorionic monoamnionic
- Dichorionic monoamnionic
- .Dichorionic diamnionic monozygotic
- Monochorionic diamnionic

**65- Human Chorionic Gonadotropin (hCG):**

- . is produced by the syncytiotrophoblast
- . enters the maternal blood circulation
- . stimulates the production of progesterone
- ✓ . A, B, and C

**66- The zona pellucida plays an important role in**

- the formation of the prochordal plate
- neural crest migration
- embryonic blood production
- ✓ preventing premature implantation

**67- Phenomena that creates female gametes, oogenesis is ceased at prophase I up till puberty. This describes the DNA content of a female's gametes during her childhood.**

- 23 chromosomes, 23 chromatids
- 23 chromosomes, 46 chromatids
- ✓ 46 chromosomes, 92 chromatids
- 46 chromosomes, 46 chromatids

**68- Cells in somites give rise to all**

- nuclei pulposi
- of the abdominal diaphragm
- bones of the skull
- ✓ skeletal muscles of the trunk

**69- The central nervous system forms in the sequence:**

- .notochord to neural plate to neural tube
- .neural tube to neural plate to neural groove
- ✓ .neural plate to neural groove to neural tube
- . plate to neural crest to neural zone

**70- Maternal serum alpha-fetoprotein increases if the fetus has**

- . oligohydramnios
- Bochdalek hernia
- ✓ anencephaly
- .meningocele

**71- Which of the following is associated with oligohydramnios**

- .tracheo-esophageal fistula
- ✓ renal agenesis
- .myelomeningocele
- .anencephaly

**72- Which process in the third week of development converts the embryonic disc into the trilaminar embryo.**

- ☐ .placentation
- ☒ .gastrulation
- ☐ .neuralation
- ☐ .somitogenesis

**73- In female child at birth , Primordial follicle contain:**

- ☒ primary oocyte arrested at prophase I
- ☐ .Secondary oocyte arested at metaphase II
- ☐ .Primordial germ cells
- ☐ .Oogonia

**74- Which of the following WILL contribute to the maternal part of the placenta?**

- ☐ , chorionic plate
- ☐ , Decidua capsularis
- ☒ decidua basalis
- ☐ , decidua marginalis

**75- The correct sequence from cranial to caudal of the secondary brain vesicles is:**

- ☐ prosencephalon,mesencephalon,metencephalon,myelencephalon, rhombencephalon
- ☐ .telencephalon,diencephalon, metencephalon, mesencephalon, myelencephalon
- ☒ .telencephalon, diencephalon, mesencephalon, metencephalon, myelencephalon
- ☐ .prosencephalon,diencephalon,mesencephalon,myelencephalon, metencephalon

**76- The unspecialized germ cells found in the seminiferous tubules are called**

- ☐ .Primary spermatocytes
- ☐ -Secondary spermatocytes
- ☒ -Spermatogonia
- ☐ .Primordial germ cells

**77- Which process establishes the 3 definitive germ layers?**

- ☐ . Neuralation
- ☒ . Gastrulation
- ☐ . Craniocaudal folding
- ☐ . Lateral folding

**78- The average duration of gestation measured from the**

**first day of the last normal menstrual period**

- ☐ 34 weeks
- ☐ .38 weeks
- ☒ .40 weeks
- ☐ .42 weeks

**79- During week 2 the blastocyst implants and the outer cell layer in that week becomes:**

- ☐ .decidual cells and hypoblast cell layer
- ☒ .cytotrophoblasts and syncytiotrophoblasts
- ☐ .chorion and amniotic membranes
- ☐ .the embryoblast and inner cell mass

**80- Between which two layers is the extraembryonic mesoderm located?**

- ☐ Epiblast and hypoblast
- ☐ . Syncytiotrophoblast and cytotrophoblast
- ☐ Syncytiotrophoblast and endometrium
- ☒ . Exocoelomic (Heuser's) membrane and cytotrophoblast

**81- The intraembryonic coelom forms within:**

- ☐ .somites
- ☒ .lateral plate
- ☐ .neural tube
- ☐ .intermediate mesoderm

**82- The process of transformation of a spermatid into a spermatozoan is called as**

- ☐ . Spermatogenesis
- ☐ .Gametogenesis
- ☒ .Spermiogenesis
- ☐ Sperm maturation

**83- Pregnancy tests generally are based upon the detection in maternal urine of:**

- . Estrogen
- . Progesterone
- . human Chorionic Somatomotropin
- ✓ . human Chorionic Gonadotrophin